Literature Review

CADepth-Net: Channel-wise Attention-based Network for Self-Supervised Monocular Depth Estimation

Existing self-supervised monocular depth estimation methods fail to explicitly model scene structure in RGB images and to preserve object boundaries .Because existing CNN-based methods implicitly learn scene structure instead of explicitly modeling it , the predicted maps tend to be less accurate and contain blurry artifacts .

To address this, the authors introduce CADepth-Net, which enhances the existing encoder-decoder architecture to better model scene structure and preserve object boundaries compared to previous methods .The proposed model combines two modules: the Structure Perception Module(SPM) which utilizes self-attention mechanism to obtain rich context of scene structure and better scene understanding, and the Detail Emphasis Module(DEM) which utilizes channel attention mechanism to fuse different scale features and emphasize important details for sharper depth estimation resulting in fewer blurry artifacts.

The model shows strong generalization on the unseen Make3D dataset, producing more accurate and sharper depth predictions than prior methods. SPM improves scene understanding with almost no added computational cost, while DEM's improved detail preservation comes with some increase in computational complexity reflecting a trade-off between accuracy and efficiency. Despite this added cost, the full model remains suitable for real-time use on high-end GPUs.

However, evaluation is limited to KITTI and Make3D, so performance in more diverse conditions (night, rain, fog, or indoor scenes) remains untested. Future work should evaluate the model on more diverse datasets and real-world driving scenarios, explore lightweight attention modules for edge devices, and investigate robustness under adverse environmental conditions.

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